

Statistical Machine Learning Report

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Abstract

Data analytics in sport provides essential tools to assess and improve athletic performance. This project presents a study on padel strokes classification using Machine Learning algorithms. Using inertial data collected from a smartphone worn on the arm, 7 representative strokes of this sport were recorded and analyzed. Through feature engineering techniques, the data were processed and fed into four predictive models (K-Nearest Neighbors, Random Forest, Support Vector Machine, and Multi-Layer Perceptron). Beyond the modeling task itself, this report documents a methodological pitfall discovered and corrected over the course of the project: an initial data collection protocol produced a severe, easy-to-miss data leakage issue, which was diagnosed and fixed by redesigning the recording protocol. The final aim is to demonstrate the effectiveness of these systems in recognizing complex biomechanical movements, offering excellent predictive accuracy with an accessible hardware setup.

1 Introduction

The central problem addressed by this project is the automatic identification of the stroke type performed by the player, translating the physical signals recorded on the arm into a multiclass classification problem, proposing a practical and efficient approach. Temporal data were acquired through the inertial sensors (IMU) of a smartphone housed in a sports armband, using the *Sensor Logger* application.

In summary, the main objectives and contributions of this project are:

- The diagnosis and correction of a severe data leakage issue introduced by a naive data collection protocol, and the redesign of that protocol around mixed-block recording sessions (Section 2).
- The application of **feature engineering** techniques to extract relevant information (such as mean, maximum, and minimum values) from the raw time series.
- The training and comparison of four Machine Learning algorithms of very different nature: **K-Nearest Neighbors (KNN)**, **Random Forest**, **Support Vector Machine (SVM)**, and **Multi-Layer Perceptron**.
- The creation of a dataset targeting the recording of 7 fundamental padel stroke types.

(MLP) to evaluate the predictive accuracy of the system in distinguishing movements.

2 Methodological Journey: From a Weak Protocol to a Robust One

Before presenting the final experimental setup, this section documents a methodological issue that shaped the entire project and that we believe is worth reporting explicitly, since it is a "trap" easy to fall into and easy to miss in a project like this one.

2.1 Initial Protocol and a Deceptively High Accuracy

The data collection process originally followed what seemed like a natural protocol: each recording session was dedicated to a single stroke type (e.g., one continuous recording containing only forehand repetitions, a separate recording containing only backhand repetitions, and so on), repeated across a small number of players and sessions. A K-Nearest Neighbors classifier ($k = 1$) was trained on a stratified random 70/30 split of individually windowed strokes, in the style common in the literature on IMU-based activity recognition.

This setup produced an accuracy at or near 100%, for the full 7-class problem as well as for challenging sub-problems such as distinguishing `smash_in` from `smash_out`. However, while these results seemed astonishing, they actually hid a fundamental problem.

2.2 Diagnosis: Data Leakage

The suspiciously perfect accuracy prompted a systematic investigation. Because every stroke of a given class in this protocol originated from a single, uninterrupted recording session, the identity of the session and the identity of the class label were perfectly confounded: knowing which session a stroke came from was equivalent to knowing its label.

Re-evaluating the classifier with `GroupKFold` cross-validation, using the recording session as the grouping variable (leave-one-session-out, LOSO), instead of a random split, caused accuracy to collapse dramatically from about 100% down to roughly 20%, close to or in some configurations below the chance level for the number of classes involved. For the `smash_in` versus `smash_out` sub-task specifically, accuracy collapsed too, suggesting, incorrectly, that the two strokes were biomechanically indistinguishable with this sensor setup.

A further diagnostic confirmed the leakage directly: removing every engineered feature sensitive to a constant offset (features summarizing the signal's mean value per axis, see Section 5) caused accuracy on the flawed dataset to drop from 32.0% to 16.6%, essentially the chance level, showing that the model's apparent skill relied almost entirely on session-specific calibration and mounting bias rather than on the shape of the movement itself.

The root cause: with one class per session, any systematic artifact of a given recording: how the phone happened to be strapped to the arm that

day, small orientation offsets, sensor calibration drift becomes a perfect, spurious predictor of the class label. A classifier evaluated with a random split can freely exploit this shortcut, since strokes from the same session end up on both sides of the split.

2.3 Redesigning the Protocol: Mixed-Block Recording Sessions

To solve this initial data collection problem, we choose to change our approach. Instead of dedicating an entire recording session to a single stroke type, each new recording session interleaves all 7 stroke types, executed in short repeated blocks (approximately 15 strokes each), separated by short pauses of 10-15 seconds marking the transition between stroke types. Under this design, any session-specific artifact is now shared across every class within that session, and can no longer act as a shortcut for classification. This means the classifier is forced to rely on genuine kinematic differences between stroke types.

As an additional safeguard, the order of stroke types was varied between recording sessions (rather than repeating the exact same sequence every time), to prevent a subtler confound in which a stroke type’s fixed position within the sequence, rather than its class, could become predictive if some signal property drifted systematically over the course of a session (e.g., due to sensor warm-up or player fatigue). A dedicated diagnostic test, predicting whether a stroke belonged to the first or second half of its session, using only strokes of a single class at a time, returned an accuracy close to chance (51.3%), confirming that no such positional artifact was

present.

2.4 Automatic Block Discovery

Manually inspecting every recording to identify block boundaries does not scale and is difficult to justify as an objective, reproducible procedure. An automatic block-discovery algorithm was therefore developed, applied uniformly to every session: every stroke impact is located via peak detection on the acceleration magnitude signal; every candidate pause threshold between consecutive strokes is tested (the midpoint between each pair of consecutive, sorted, inter-stroke time gaps); thresholds that do not produce a number of blocks that is a whole multiple of the 7 stroke types in the protocol are discarded, since a valid recording always consists of complete laps of the base sequence (the 7 strokes); among the remaining thresholds, the one producing the most uniform block sizes is selected (lowest coefficient of variation of strokes-per-block).

This procedure determines block boundaries systematically rather than through manual, case-by-case inspection, and was verified to reproduce, from raw signal data alone, boundaries matching those previously identified by hand.

3 Data Collection and Experimental Setup

To ensure the robustness of the Machine Learning model and its ability to generalize across different movements, data collection involved 2 players. The biomechanical variability in-

troduced by the different playing styles and body builds of the participants is indeed a fundamental element to prevent the model from overfitting to a single user. The experimental setup involved a smartphone firmly attached to the player’s dominant arm via an armband. This strategic positioning did not interfere with the player’s freedom of movement during stroke execution, while ensuring a faithful measurement of wrist and forearm dynamics.



Figure 1: Armband setup on the arm.



Figure 2: The four project participants on a padel court during a data acquisition session.

To cover the most common and funda-

mental dynamics of a padel match, the dataset was built by recording 7 specific stroke types: Forehand, Backhand, Lob, Smash in, Smash out, Forehand Volley, and Backhand Volley (see Figure 1 for the acquisition setup), following the mixed-block protocol described in Section 2.3.

Continuous signal recording was performed using the mobile application *Sensor Logger*. Participants repeatedly performed the target strokes, and the application was configured to sample exclusively the raw data from the inertial sensors:

- **Accelerometer (Linear Acceleration):** detects linear acceleration along the X, Y, Z axes.
- **Gyroscope (Rotation Rate):** tracks angular velocity and wrist rotations.
- **Gravity:** isolates the gravitational component of acceleration, revealing the device’s orientation independent of movement.

The final dataset comprises 6 recording sessions across the 2 players, segmented automatically into labeled strokes using the block-discovery procedure described in Section 2.4.

4 Preprocessing and Data Cleaning

The raw data extracted from the smartphone consist of two separate time streams: accelerometer and gyroscope readings. The first preprocessing step involved synchronizing these two signals based on the nearest timestamp, obtaining a single time-aligned dataset. Subsequently, to simplify movement

identification, the overall acceleration magnitude was computed, defined as the Euclidean norm of the vector:

$$acc_mag = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

To isolate individual strokes within the continuous recording, a peak detection algorithm was applied based on the acceleration magnitude. Once each stroke’s peak was located, the data were cropped into fixed time windows of 50 samples centered on the peak itself, ensuring a uniform input length.

Finally, to ensure the highest dataset quality, rigorous outlier cleaning was implemented. Instead of classical standardization, which is sensitive to extreme outliers, a robust Z-score based on the Median Absolute Deviation (MAD) was employed, computed as follows:

$$MAD = \text{median}(|x_i - \tilde{x}|)$$

$$Z_{robust} = 0.6745 \cdot \frac{x_i - \tilde{x}}{MAD}$$

where $\tilde{x}$ represents the sample median. Any stroke with a Z_{robust} value above the threshold of 3.5 was discarded, ensuring that the model was trained only on biomechanically consistent executions.

5 Feature Engineering

Rather than feeding the model raw point-by-point time series, an approach based on characteristic extraction (*Feature Engineering*) was adopted, to synthesize the dynamic behavior of each stroke into meaningful metrics.

For each time window corresponding to a single stroke, statistical and physical metrics were extracted from the 6

available channels (acceleration and gyroscope along the X, Y, and Z axes). Specifically, for each axis the following were computed:

- **Standard Deviation and Range:** to quantify the variation, explosiveness, and total amplitude of the arm’s oscillation.
- **Mean:** to identify the baseline orientation and directional bias of the stroke.
- **Energy:** useful to capture the overall intensity of the movement, computed as the mean squared error with respect to the signal’s mean:

$$E = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2$$

In addition, global features based on the acceleration magnitude were extracted, such as the absolute peak value and the relative temporal position of the peak within the window. This process transformed complex signals into a set of structured variables, ideal for training traditional classifiers efficiently. At the end of this feature engineering step, we ended up with 28 features per stroke in total.

6 Exploratory Data Analysis (EDA)

Before proceeding with model training, an exploratory analysis of the clean dataset was conducted, with the goal of verifying class balance, the quality of the engineered features, and the presence of distinctive patterns across stroke types.

6.1 Class Balance

Figure 3 shows the distribution of the number of strokes for each of the 7 classes after outlier removal.

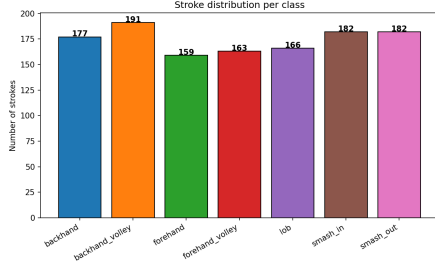


Figure 3: Stroke distribution per class.

The dataset is reasonably balanced: the number of strokes per class ranges from a minimum of 159 (forehand) to a maximum of 191 (backhand_volley), with a relative difference contained within 20%. This justifies the use of balanced accuracy as the optimization metric (Section 7.1), more as methodological caution than as a necessity imposed by a severe imbalance, but it does exclude the need for resampling techniques (*oversampling/undersampling*).

6.2 Distribution of Key Features per Class

Figure 4 reports boxplots of three particularly informative features: the peak and energy of the acceleration magnitude (`acc_mag_max`, `acc_mag_energy`) and the gyroscope range on the X axis (`x_gyro_range`).

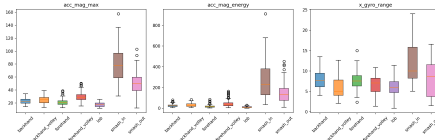


Figure 4: Boxplot of three key features per class.

A clear intensity hierarchy emerges among the strokes: `smash_in` stands out sharply from the other classes with much higher peak and energy values (median peak ≈ 78 , against values between 17 and 27 for forehand, lob, backhand, and volleys), followed by `smash_out` with intermediate values. The remaining classes (forehand, lob, backhand, forehand_volley, backhand_volley) instead show overlapping distributions and similar medians, which anticipates the classification difficulty observed later in distinguishing these classes from one another (see the confusion matrix in Section 8.2). A residual of outliers is also visible for almost all classes: this is expected, since outlier removal in Section 4 was performed on a different (and broader) set of features than the three visualized individually here.

6.3 Feature Correlation

Figure 5 shows the correlation matrix across all 28 engineered features.

Two main patterns emerge. First, for each channel, the standard deviation, range, and energy features are strongly correlated with one another, probably because they all capture, in slightly different ways, the amplitude of the signal’s oscillation. Second, the **mean** features per channel (`_mean`) behave qualitatively differently from the rest: they show weak or negative correlations both with other mean features and with the variation features. This confirms that the mean features carry complementary, non-redundant information relative to the rest of the set, consistent with the sharp accuracy drop observed when these features are removed on the “weak” dataset (Section 2.2).

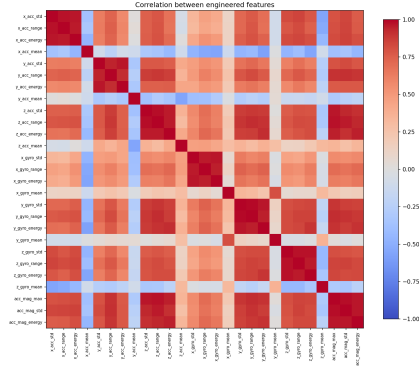


Figure 5: Correlation matrix between the engineered features.

6.4 Intra-Class Variability

To assess how consistent the signal shape is within the same class, Figure 6 overlays 8 example `acc_mag` signals for each stroke type.

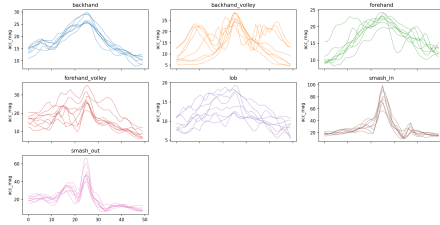


Figure 6: Intra-class variability: 8 overlaid strokes per type.

Forehand, backhand, and `smash_in` show a very consistent waveform across repetitions, with the peak aligned almost always at the same point in the window: this explains why these classes obtain the best performance in the final model (Table 2). `Smash_out` presents a characteristic double peak (a smaller preliminary oscillation followed by the main peak), reproduced fairly systematically across repetitions. Lob, forehand_volley, and backhand_volley in-

stead show greater irregularity, with more jagged waveforms and secondary peaks of variable amplitude from repetition to repetition: this greater intra-class variability is consistent with the lower F1-score obtained by these classes, in particular forehand_volley, in Section 8.2. This fact is also due to the probability that the way of executing these shots differs significantly between the two people who recorded them.

6.5 Class Separation: Peak vs Energy

Figure 7 reports a 2D scatter plot between the peak and the energy of the acceleration magnitude, colored by class.

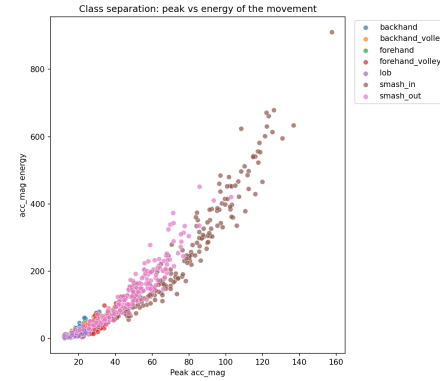


Figure 7: Class separation in the peak vs energy feature space.

The two variables are strongly correlated with one another (an almost linear relationship), as expected given that energy is computed from deviations around the signal's mean value, of which the peak is the dominant component. Along this relationship, `smash_in` separates clearly from the other classes, occupying the region of higher peak and energy values, with `smash_out` in an intermediate position. The remaining five classes (forehand, lob, backhand,

forehand_volley, backhand_volley) are instead strongly overlapping in this two-dimensional projection, confirming that two features alone are not sufficient to separate them: their distinction necessarily requires the full 28-dimensional feature space used by the models (Section 7).

6.6 Stroke Distribution per Person

Finally, Figure 8 verifies that class balance is maintained at the level of individual participants.

Both participants contributed a comparable number of strokes for each class, without significant systematic imbalances: the widest gaps are observed for forehand and backhand_volley, but remain contained relative to the total. This balance between people is a useful precondition to ensure the models learn to generalize across different people.



Figure 8: Stroke distribution per person and class.

7 Classification Models

7.1 Validation Scheme

Model comparison was conducted at two levels.

As a first reference, a stratified 70/30

random split was performed on the raw time series, using a KNN with $k = 1$ as a purely indicative baseline, in the style common in the literature. This step is not used to select the final model: since strokes from the same session may share patterns tied to the specific arm-band positioning of that session, a high accuracy here does not guarantee any real generalization, in fact this is precisely the leakage risk documented in Section 2.

The validation actually used to compare and select models is leave-one-session-out (LOSO), implemented with **GroupKFold** using the recording session as the group: at each iteration an entire session is excluded from training and used as the test set, repeating the procedure for all available sessions. Predictions obtained on each excluded (out-of-fold) session are concatenated into a single vector, used to compute the accuracy, balanced accuracy, and classification report reported in Section 8.

The same LOSO scheme was reapplied, separately, to a sub-task restricted to the smash_in and smash_out classes only (Section 7.3), to verify whether our models were able to detect the differences in the same stroke type that cause the ball to go in or out.

In all comparisons the selection metric is balanced accuracy rather than plain accuracy, to avoid favoring models that merely predict the more represented classes well (Section 6.1).

7.2 Model Comparison

Four models were compared, each using a hyperparameter grid explored via **GridSearchCV** (Table 1).

Model	Hyperparameters explored
KNN	$k \in \{1, 3, 5, 7\}$, $p \in \{1, 2\}$
Random Forest	$n_estimators \in \{100, 200\}$ $max_depth \in \{None, 10\}$
SVM	$C \in \{0.1, 1, 10, 100\}$ $kernel \in \{rbf, linear\}$ $\gamma \in \{scale, auto\}$
MLP	$hidden_layers \in \{(32), (64), (32, 16)\}$ $\alpha \in \{0.0001, 0.001, 0.01\}$ $activation \in \{relu, tanh\}$

Table 1: Hyperparameter grids explored for each model.

For SVM and MLP, sensitive to feature scale, the model was wrapped in a `Pipeline` with `StandardScaler` upstream; for KNN and Random Forest, standardization was not applied.

Hyperparameter search was not performed on a separate split, but **inside** the leave-one-session-out scheme: each configuration is evaluated on the average balanced accuracy obtained by excluding each session in turn. This avoids selecting a configuration that fits a single random split well but does not hold up on unseen recordings.

The best configuration for each model was then reused to regenerate out-of-fold predictions on all sessions, used for the final comparison of the four algorithms in Section 8. The model with the highest LOSO balanced accuracy was chosen as the reference for subsequent analyses (confusion matrix, per-class metrics, PCA).

7.3 Sub-task: `smash_in` vs `smash_out`

The `smash_in` and `smash_out` classes were isolated from the rest of the dataset and subjected to the same four-model comparison, with the same LOSO scheme and the same hyperparameter grids from Table 1. The goal was to verify whether, once the initial problem of data leakage was resolved,

the models were capable of accurately solving a binary classification task between `smash_in` and `smash_out`, which means correctly predicting whether a ball hit with the same type of stroke is in or out.

8 Results

8.1 Model Comparison (Leave-One-Session-Out)

Table 2 reports accuracy and balanced accuracy of the four models under the leave-one-session-out scheme, with the respective hyperparameters selected by `GridSearchCV`.

Model	Accuracy	Bal. Acc.
KNN	83.4%	83.0%
Random Forest	87.9%	87.5%
SVM	94.0%	93.9%
MLP	93.1%	92.9%

Table 2: Comparison of the four models, LOSO scheme (6 sessions), 7 classes (chance level 14.3%).

The best model is SVM (Support Vector Machine), with an RBF kernel, $C = 100$ and $\gamma = \text{scale}$, closely followed by MLP (tanh activation, $\alpha = 0.0001$, single hidden layer with 32 units). KNN and Random Forest remain 6–10 percentage points behind.

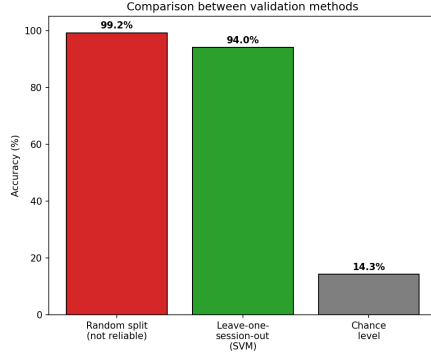


Figure 9: Comparison between stratified random split, leave-one-session-out (SVM), and chance level.

Figure 9 compares SVM’s performance in leave-one-session-out with that obtained via the stratified random split (Section 7.1). The stark gap between the two accuracies is the same signature of data leakage diagnosed and corrected in Section 2: under a random split, strokes from the same session — sharing correlated signal properties, such as armband positioning or session-specific recording conditions, can be split between training and test sets, letting the model exploit such regularities and obtain an artificially high score. The drop observed under leave-one-session-out therefore does not represent a deterioration, but a more rigorous estimate of the performance expected on unseen recording sessions. This is exactly what a classification model should be able to do, that is to behave well with new unseen registrations, so that one can analyze, for example, the difference in the performances of a player in different workout sessions.

8.2 Analysis of the Best Model

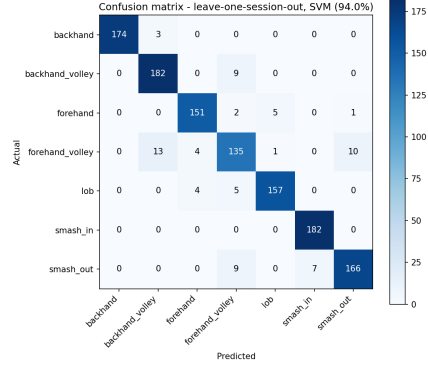


Figure 10: Confusion matrix, SVM, leave-one-session-out (94.0%).

Figure 10 shows the SVM confusion matrix on out-of-fold predictions (1220 strokes in total). The diagonal is clearly dominant for all classes, but two recurring error patterns emerge:

- **forehand_volley** is the most critical class, confused 13 times with **backhand_volley** and 10 times with **smash_out**; it is also the only class with an F1-score below 0.90 (0.836, Figure 11).
- **smash_out** is confused 9 times with **forehand_volley** and 7 times with **smash_in**, a sign that the intensity of a mis-hit (out) smash can sometimes overlap with both short net volleys and a valid smash.

The **backhand** and **smash_in** classes are instead nearly perfectly separated, with only 3 errors out of 177 and 0 errors out of 182, respectively.

Figure 11 reports precision, recall, and F1-score for each class. Backhand and smash_in reach $F1 \geq 0.98$; forehand_volley remains the weakest class

(precision 0.844, recall 0.828), consistent with the double source of confusion just described.

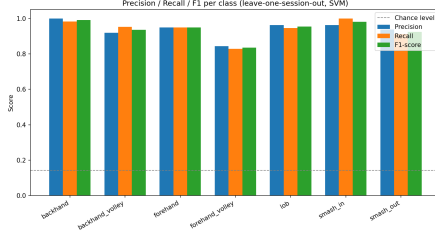


Figure 11: Precision, recall, and F1-score per class, SVM, leave-one-session-out.

8.3 Class Separability

Figure 12 shows the projection of the 28 engineered features onto the first two principal components, which together explain 73.1% of the total variance. Smash_in forms a clearly separated cluster along the first component, consistent with its markedly higher impact energy (Section 6.2). The remaining classes instead occupy a compact, partially overlapping region near the origin. Smash_out sits in an intermediate position, partially overlapping both the central cluster and smash_in, consistent with the errors observed in the confusion matrix (Figure 10).

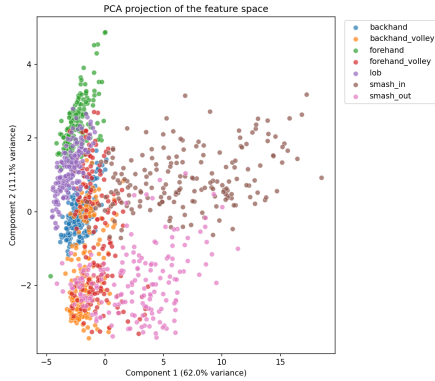


Figure 12: PCA projection of the feature space (28 dimensions $\rightarrow$ 2 components).

8.4 Sub-task: smash_in vs smash_out

Isolating the two smash classes alone (364 strokes, 182 per class), results improve markedly for every model compared to the multiclass problem (Table 3), confirming that the two classes are genuinely separable once the data leakage identified in Section 2 is removed.

However, to be fair, it is worth noting that these results are the consequence of our exaggeration when executing the two types of strokes. In fact, to perform the smash_in, we attempted to snap the wrist as much as possible so that the ball would bounce as close to the net as possible. As for the smash_out, the wrist was not snapped at all, causing the ball to hit the back glass directly.

Model	Accuracy	Bal. Acc.
KNN	96.2%	96.2%
Random Forest	97.5%	97.5%
SVM	99.5%	99.5%
MLP	99.7%	99.7%

Table 3: Comparison of the four models, smash_in vs smash_out sub-task, LOSO scheme (chance level 50.0%).

The best model in this case is MLP, with only a single error out of 364 strokes (precision and recall between 0.995–1.000 on both classes).

9 Conclusion

This project set out to classify padel strokes from forearm-worn IMU data, and delivers a system that does so

with high accuracy (94.0% across 7 classes under rigorous leave-one-session-out validation, chance level 14.3%) using a Support Vector Machine on 28 engineered features. Just as importantly, the project documents a complete methodological cycle: an initial data collection protocol produced a deceptively high but unreliable accuracy due to data leakage between session identity and class label (Section 2); the issue was diagnosed through group-aware cross-validation and a targeted feature-ablation test; and the protocol was redesigned around mixed-block recording sessions with an automatic, reproducible block-discovery procedure, eliminating the confound at its source rather than compensating for it after the fact.

The dedicated sub-task for the binary smash_in and smash_out problem was developed with the aim of understand whether our features and models were capable of distinguish two different outcomes of the same stroke. However, as previously said, the results are not very honest since we really exaggerated the difference in the two movements during registration.

Future work could extend the dataset to more players to obtain a stable estimate of cross-person generalization, and it would be interest to test the robust strength of these models in distinguishing different outcomes of the same stroke executed in a more natural manner.

References

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